Personal Loan Purchase Prediction

AllLife Bank: Machine Learning

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Executive Summary

- We have been able to build a predictive model:
  - The bank can deploy to identify customers who will be interested in taking a personal loan.
  - The bank can use to find the key factors that will have an impact on a customer taking a personal loan or not.
- Bank can look into important attributes that have an impact on Personal_loan, and they are
  - Income
  - Family
  - Education
- Higher income customers should be the target customers for the bank - Customers who have income above 116k dollars and a family of more than 2, such customers have higher chances of taking personal loans.

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Executive Summary

- Higher education higher are the chances to take a loan - Customers who are more educated (education level greater than undergraduate) have a higher chance of taking a personal loan.
- Size of the family has a positive correlation with the personal loan, as the size of the family increases (generally a family size of 3 or more than 3 members) have more chances of a customer taking a personal loan.
- Our analysis showed that ~50% of the customers who have the certificate of deposit with the bank (CD_Account) are the ones that have the requirement of Personal Loan - Bank should target such customers.

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Business Problem Overview and Solution Approach

- AllLife Bank is a US bank that has a growing customer base. The majority of these customers are liability customers (depositors) with varying sizes of deposits. The number of customers who are also borrowers (asset customers) is quite small, and the bank is interested in expanding this base rapidly to bring in more loan business and in the process, earn more through the interest on loans. In particular, the management wants to explore ways of converting its liability customers to personal loan customers (while retaining them as depositors).
- A campaign that the bank ran last year for liability customers showed a healthy conversion rate of over 9% success. This has encouraged the retail marketing department to devise campaigns with better target marketing to increase the success ratio.
- The task is to analyze the data provided by AllLife bank and build a model that will help the marketing department to identify the potential customers who have a higher probability of purchasing the loan.

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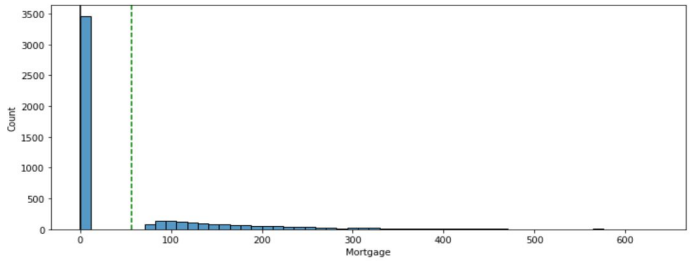
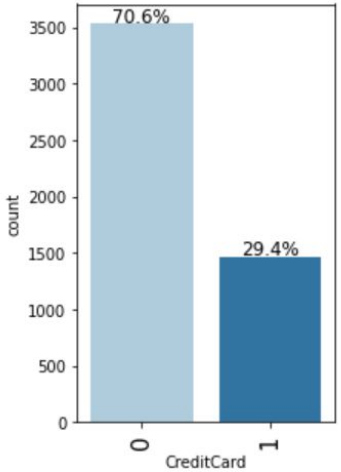
EDA Results

- Most customers have not mortgaged their house but there are many outliers.
- Some customers have a mortgage house value of more than 600k dollars.
- Such customers can be potential customers who require a personal loan.



- Approximately 70% of customers do not have a credit card issued by any other bank.



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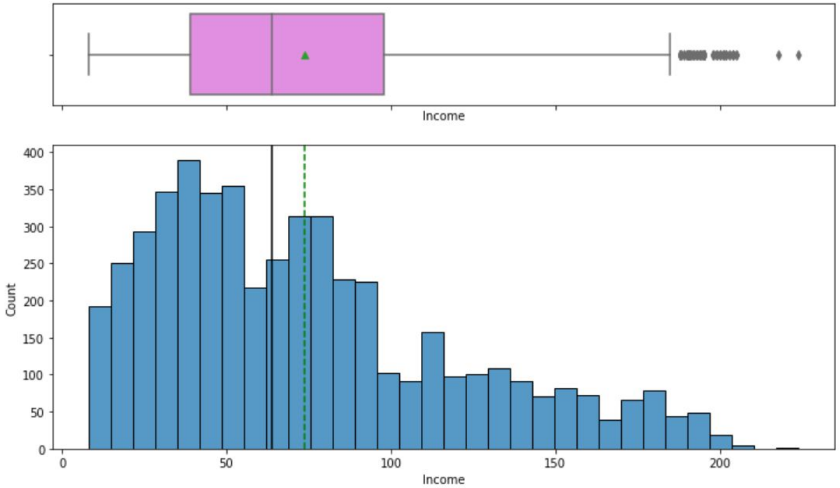
EDA Results



- The distribution of Income is skewed to right.
- Some customers have an Income above 200k dollars.
- 50% of the customers have income less than 64k dollars.

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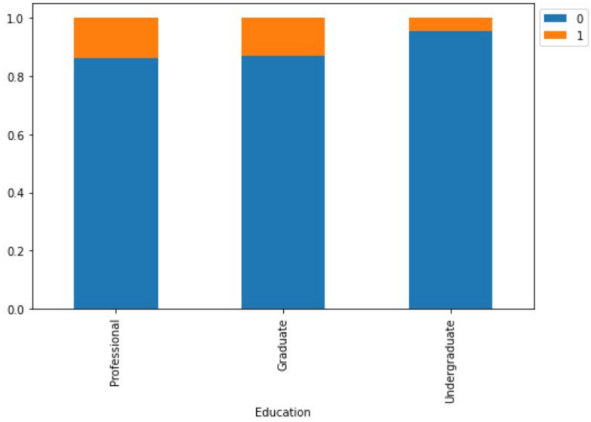
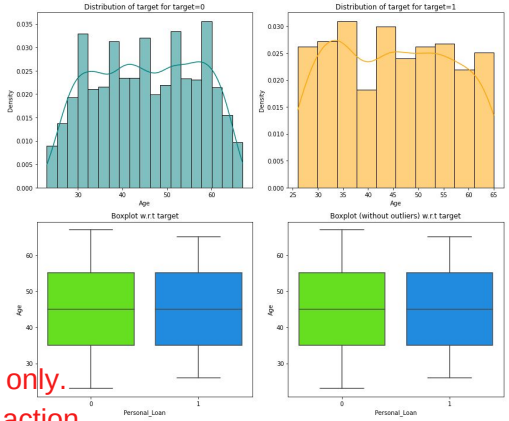
EDA Results

- ~15% of the customers who have done advanced/professional studies are the ones that require a personal loan.
- Undergraduates have the least requirement of personal loans.



- The customers who have the requirement of a Personal Loan have a wider range than the ones who do not require a Personal Loan



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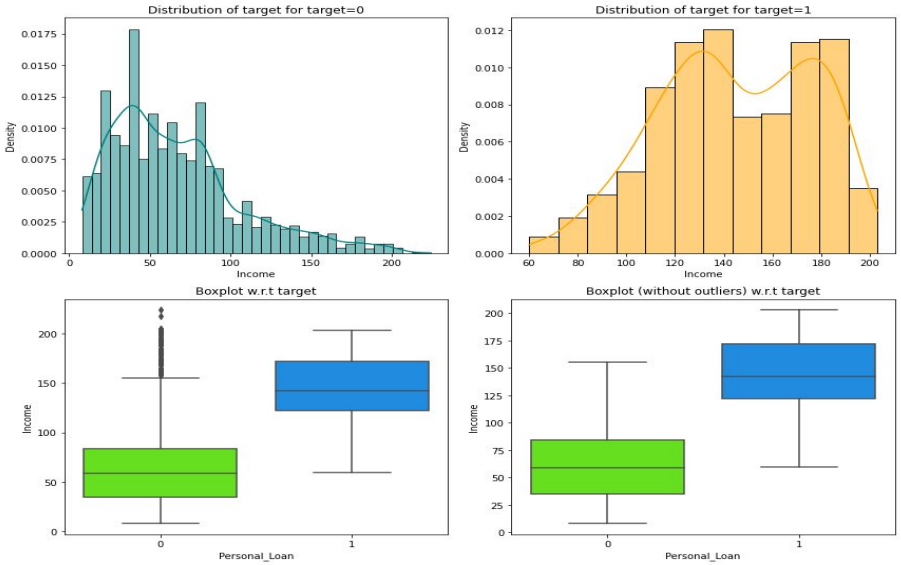
EDA Results



- Those customers who have an income higher than 90k-100k dollars are the potential customers who will take the Personal Loan.
- Income seems to be a significant predictor as it provides a good separation between two classes.

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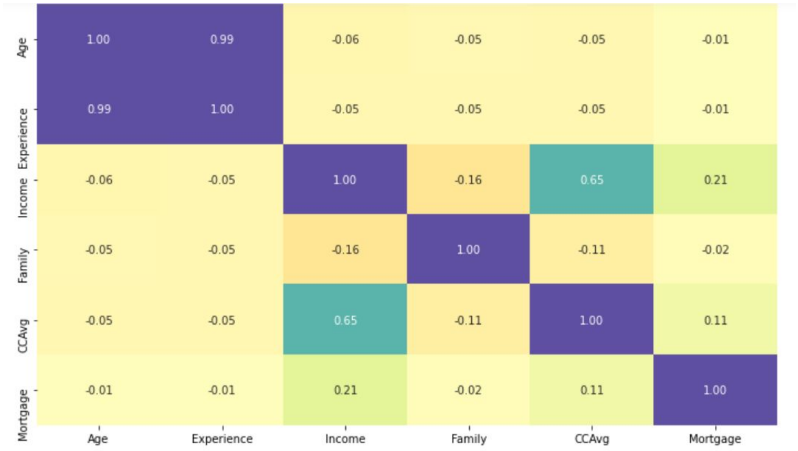
EDA Results



- Age and Experience have a perfect correlation hence one of these variables can be dropped while model building as they will provide the same information to the model.
- Income and CCAvg have a moderate correlation which makes sense as the Income increases the spendings might also increase.
- Family has a negative correlation with Income, which is quite surprising as family size increases the income of the family decreases. But this correlation is not too strong to conclude.

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Data Preprocessing

- There are 467 unique values in the zip code. In the US, The first digit of a PIN indicates the zone or a region, the second indicates the sub-zone, and the third, combined with the first two, indicates the sorting district within that zone. The final three digits are assigned to individual post offices within the sorting district
- So, zip codes are grouped based on the first 2 digits to reduce the number of unique values in the ZIPCode column.
- There are negative values in Experience and we assume that these negative signs here are data input errors, so we have replaced them with positive values.
- There are 6 categorical independent variables but 4 of them are binary, so we'll have the same results with them even after creating dummies.

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Model Building

- We want to predict whether a liability customer will buy personal loan or not using the information provided to us.
- We will use the Recall as the performance metric for our model because
  - Predicting a customer will take the personal loan, but in reality, the customer will not take the personal loan - Loss of resources
  - Predicting a customer will not take the personal loan, but in reality, the customer was going to take the personal loan - Loss of opportunity
  - We would want Recall to be maximized. The greater the Recall higher the chances of minimizing false negatives

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Model Performance Summary

- We have built the Decision Tree from sklearn library with default parameters and below are the metrics on train and test data.

	Accuracy	Recall	Precision	F1_Score
Train	1.0	1.0	1.0	1.0
Test	0.981	0.899	0.911	0.905

- The Decision Tree model indicates that the most significant predictors of buying a personal loan:
  - Education Undergraduate
  - Income
  - Family
  - CCAvg

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Model Performance Improvement: Pre-pruning

- The decision tree model with default parameters is overfitting the training data and is not able to generalize well.
- We tested the following hyperparameters using GridSearch to get the optimal model
  - max_depth
  - min_samples_leaf
  - max_leaf_nodes

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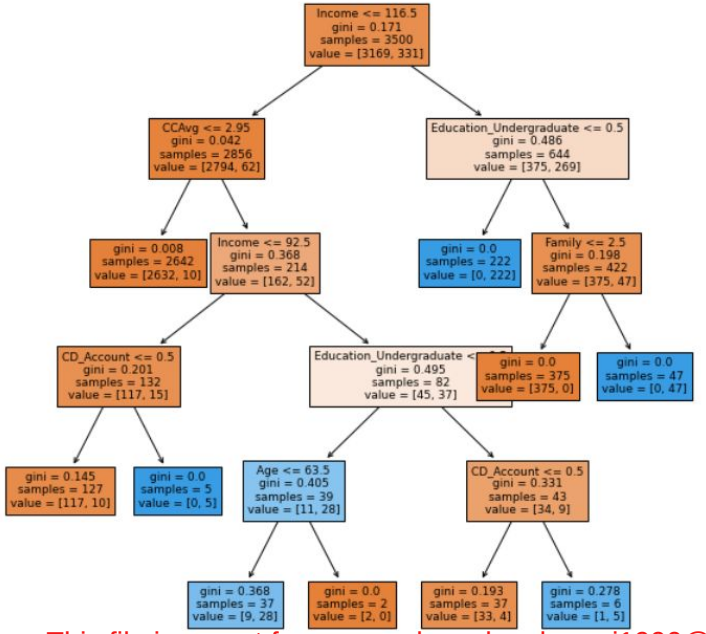
Model Performance Improvement: Pre-pruning

- Model performance improvement using Pre-Pruning
- Visualization of pre pruned tree is as follows



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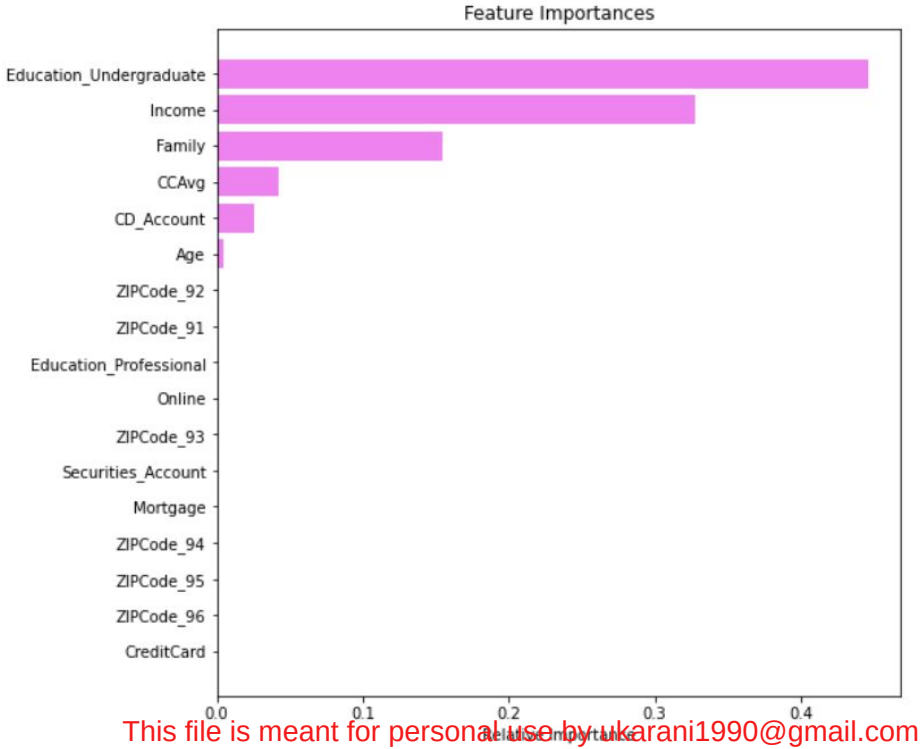
Model Performance Improvement: Pre-pruning

- Feature Importance of Decision Tree with Pre-Pruning



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Model Performance Improvement: Pre-pruning

	Decision Tree sklearn	Decision Tree (Pre-Pruning)
Accuracy	1.0	0.990286
Recall	1.0	0.927492
Precision	1.0	0.968454
F1	1.0	0.947531

Test set performance comparison:

	Decision Tree sklearn	Decision Tree (Pre-Pruning)
Accuracy	0.981333	0.980000
Recall	0.899329	0.865772
Precision	0.911565	0.928058
F1	0.905405	0.895833

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Model Performance Improvement: Post-pruning

- To overcome the overfitting problem of the default decision tree model, we tried doing cost-complexity pruning
- The best model obtained using cost-complexity pruning was the same as the default model

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Model Performance Comparison

- The pre-pruned tree has given a generalized performance with balanced values of precision and recall.

Model	Train Accuracy	Test Accuracy	Train Recall	Test Recall	Train Precision	Test Precision
Decision Tree	1	0.981	1	0.899	1	0.911
Decision Tree - Pre-Pruning	0.990	0.980	0.927	0.865	0.968	0.928

Best Performing Model

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APPENDIX

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Data Background and Contents

- The data contains information about 5000 customers.
- The attributes include Age, Experience, Income, Family, CCAvg, Education, Mortgage, and more.
- Average age of customers is 45 years, age of customers has a wide range from 23 to 67 years. The distribution of Age is fairly symmetrical about the mean and the median.
- The Experience variable seems to be fairly symmetrical about mean and median. The distribution of Experience looks like the distribution of Age. The mean and median experience of customers is equal to ~20 years.
- Average income of customers is 73k dollars. Income has a wide range from 8k dollars to 224k dollars. The distribution of Income is skewed to right.
- Most of the customers are single/live alone (~30%).
- Some customers have average spending of 0 dollars a month. The distribution of CCAvg is skewed to the right and there are many outliers.

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